Week 12 lecture 3 STAT2004/2904/7004

- Ass 4 due Fri 30 Oct @ 4pm
- Ross' revision session: Fri 30 Oct @ 2pm
  Alan's revision session: SWOTVAC Thu 5 Nov @ 10am

Recap:

① A linear model covers any model of the form:

$\underset{\sim}{Y} = X\underset{\sim}{\beta} + \underset{\sim}{\varepsilon}$, where $\underset{\sim}{Y}$ is a $n \times 1$ vector of responses
X is a $n \times p$ matrix of predictors
$\underset{\sim}{\beta}$ is a $p \times 1$ vector of parameters
& $\underset{\sim}{\varepsilon}$ is a $n \times 1$ vector of errors

② One-way (and two-way) ANOVA are examples of a linear model:

$Y_{ji} = \mu + \alpha_j + \varepsilon_{ji}$ can be written as:
(with $\alpha_1 \equiv 0$)

$$\underset{\sim}{Y} = \begin{pmatrix} Y_{11} \\ Y_{12} \\ \vdots \\ Y_{1n_1} \\ \hline Y_{21} \\ Y_{22} \\ \vdots \\ Y_{2n_2} \\ \hline \vdots \\ \hline Y_{J1} \\ Y_{J2} \\ \vdots \\ Y_{Jn_J} \end{pmatrix} \quad X = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 0 \\ \vdots & & & \\ 1 & 0 & \cdots & 0 \\ \hline 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ & & \vdots & & \\ 1 & 1 & 0 & \cdots & 0 \\ \hline & & \vdots & & \\ \hline 1 & 0 & \cdots & 0 & 1 \\ 1 & 0 & \cdots & 0 & 1 \\ & & \vdots & & \\ 1 & 0 & \cdots & 0 & 1 \end{pmatrix} \quad \underset{\sim}{\beta} = \begin{pmatrix} \mu \\ \alpha_2 \\ \alpha_3 \\ \vdots \\ \alpha_J \end{pmatrix}$$

design matrix

For the usual ANOVA null hypothesis, in the contrast parameterization this is

$$H_0: \alpha_2 = 0 = \alpha_3 = \alpha_4 = \dots = \alpha_J$$

Note that we can write this null hypothesis in matrix form:

$$H_0: R\underset{\sim}{\beta} = \underset{\sim}{0} \quad \text{versus} \quad H_0: R\underset{\sim}{\beta} \neq \underset{\sim}{0}$$

where $R = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & 0 \\ 0 & \cdots & & 0 & 1 \end{pmatrix}$ & $\underset{\sim}{\beta} = \begin{pmatrix} \mu \\ \alpha_2 \\ \alpha_3 \\ \alpha_J \end{pmatrix}$

Here, R is called a "restriction matrix" that is applied to our parameters $\underset{\sim}{\beta}$

This restriction matrix can be chosen to reflect different hypotheses about our groups. For example, in a tooth growing experiment with six groups, we may be interested in

$$H_0: \left\{ \underbrace{\mu_1 = \mu_2 = \mu_3}_{\text{OJ group}} \quad \text{and} \quad \underbrace{\mu_4 = \mu_5 = \mu_6}_{\text{tablet}} \right\}$$

"dosage doesn't matter, but there is a difference in the type of supplement"

$$\Leftrightarrow H_0: \left\{ \begin{matrix} \alpha_2 = 0 \\ \alpha_3 = 0 \end{matrix} , \quad \alpha_4 = \alpha_5 = \alpha_6 \right\}$$

$$\Leftrightarrow H_0: \left\{ \begin{matrix} \alpha_2 = 0 \\ \alpha_3 = 0 \end{matrix} , \quad \begin{matrix} \alpha_4 - \alpha_5 = 0 \\ \alpha_5 - \alpha_6 = 0 \end{matrix} \right\}$$

$$\Leftrightarrow H_0: R\underset{\sim}{\beta} = 0 \quad \text{where}$$

In general, we cannot use this statistic yet because σ^2 is itself unknown.

BUT, we can estimate σ^2 via

$$\hat{\sigma}^2 = \frac{RSS_1}{n-p} = \frac{RSS_1}{df_1} = MS_{error} = MS_{resid}$$

Moreover, $\frac{RSS_1}{n-p}$ is indep of $\frac{RSS_0 - RSS_1}{r}$

And so,

$$\frac{(RSS_0 - RSS_1)/r}{RSS_1/n-p} \sim \frac{\chi^2_r/r}{\chi^2_{n-p}/n-p} \quad \text{indep}$$

$$=: F_{r,n-p}$$

$$= F_{r,df_1}$$

So, we can summarize all of this with a linear-model F-table:

Model		df	RSS	F	$P_r(F)$
"Comp"	<difference>	r	$RSS_0 - RSS_1$	$\frac{(RSS_0 - RSS_1)/r}{RSS_1/n-p}$	$P(F_{r,n-p} \geq F)$
"resid"	alternative	$n-p$	RSS_1		
"Total"	Null	$n-(p-r)$	RSS_0		

Example 8.8 (Jump distance vs height of students from STAT1201 in 2021)

Consider the cubic model for jump distance vs. height:

$$jump_i = \beta_0 + \beta_1 height_i + \beta_2 height_i^2 + \beta_3 height_i^3 + \varepsilon_i$$

Is this a linear model?

If $\underset{\sim}{Y} = \begin{pmatrix} jump_1 \\ jump_2 \\ \vdots \\ jump_n \end{pmatrix}$ $X = \begin{pmatrix} 1 & height_1 & height_1^2 & height_1^3 \\ 1 & height_2 & height_2^2 & height_2^3 \\ \vdots & & & \\ 1 & height_n & height_n^2 & height_n^3 \end{pmatrix}$ $\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}$

then, $\underset{\sim}{Y} = X\underset{\sim}{\beta} + \underset{\sim}{\varepsilon}$

So, this IS a linear model!

So, a linear model is linear in its parameters $\underset{\sim}{\beta}$

(Aside: If $jump_i = \beta_0 + \beta_1 height_i + \beta_2 weight_i + \beta_3 BMI_i + \varepsilon_i$

$BMI_i \simeq \frac{weight_i}{height_i}$.

then $jump_i = \beta_0 + \beta_1 height_i + \beta_2 weight_i + \beta_3 \frac{weight_i}{height_i} + \varepsilon_i$)

Q: Do we really need a cubic model in height to explain jump distance, or is a quadratic (or even a linear) model sufficient?

RDEMO

$$R = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{pmatrix} \qquad \beta = \begin{pmatrix} \mu \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \\ \alpha_6 \end{pmatrix}$$

The reason why linear models are so popular is that there is a universal test for any restrictions R on the model parameters. In general, for testing

$H_0: R\beta = 0$ vs $H_1: R\beta \neq 0$ for any restriction matrix R,

we simply { fit the null model imposing the restriction
fit the alternative model without the restriction,

& then we compare the residual sum-of-squares leftover from our two fitted models:

$$\frac{(RSS_0 - RSS_1)/r}{RSS_1/df_1}, \qquad r = \#\text{ restrictions under } H_0 = \text{rank}(R) = df_0 - df_1$$

→ how much worse is the null model fit compared to the alt. model fit

RSS_1/df_1: how well does the alt. model fit the data.

$$\sim F_{r, df_1}$$

This is a generalization of ANOVA to any linear model.

Where does this universal test come from?
Under the working assumption that $\varepsilon \overset{iid}{\sim} N(0, \sigma^2)$ with σ^2 known for now.

Then

$$Y \sim N_{n\times 1}(X\beta, \sigma^2 I)$$

where $\sigma^2 I = \begin{pmatrix} \sigma^2 & & 0 \\ & \sigma^2 & \\ 0 & & \ddots \\ & & & \sigma^2 \end{pmatrix}$

The likelihood for β:

$$L(\beta \mid Y, X) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left\{-\frac{1}{2\sigma^2}(Y - X\beta)^T(Y - X\beta)\right\}$$

"$(y-\mu)^2$"

And so, under the null $H_0: R\beta = 0$, we may maximize $L(\beta \mid Y, X)$ imposing restriction $R\beta = 0$ to get

$\hat{\beta}_{\text{restricted}}$

& the maximal likelihood achieved is

$$L(\hat{\beta}_{\text{restricted}} \mid Y, X) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left\{-\frac{1}{2\sigma^2}(Y - X\hat{\beta}_{\text{restricted}})^T (Y - X\hat{\beta}_{\text{restricted}})\right\}$$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left\{-\frac{1}{2\sigma^2} RSS_0\right\}$$

Similarly, under no restrictions, we have $\hat{\beta}_{MLE}$ (or $\hat{\beta}$)

$$L(\hat{\beta} \mid Y, X) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) \exp\left\{-\frac{1}{2\sigma^2}(Y - X\hat{\beta})^T(Y - X\hat{\beta})\right\}$$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) \exp\left\{-\frac{1}{2\sigma^2} RSS_1\right\}$$

So the GLRT for testing $H_0: R\beta = 0$ vs $H_1: R\beta \neq 0$ is

$$\Lambda = \exp\left\{-\frac{1}{2\sigma^2}(RSS_0 - RSS_1)\right\}$$

Then small values of Λ ($\Leftrightarrow$ large values of $RSS_0 - RSS_1$,

where

$$RSS_0 - RSS_1 \sim \sigma^2 \chi^2_r, \text{ where } r = \text{rank}(R) = \#\text{restrictions}$$

$$\Leftrightarrow \frac{1}{\sigma^2} \frac{RSS_0 - RSS_1}{r} \sim \chi^2_r / r$$